

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH  
MERN

# Proposed Solution

ShopEZ — MERN Stack E-commerce Application

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**Phase:** Project Design Phase

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**Classification:** Project Documentation — For Academic Submission

# Proposed Solution

## 1. System Context Diagram

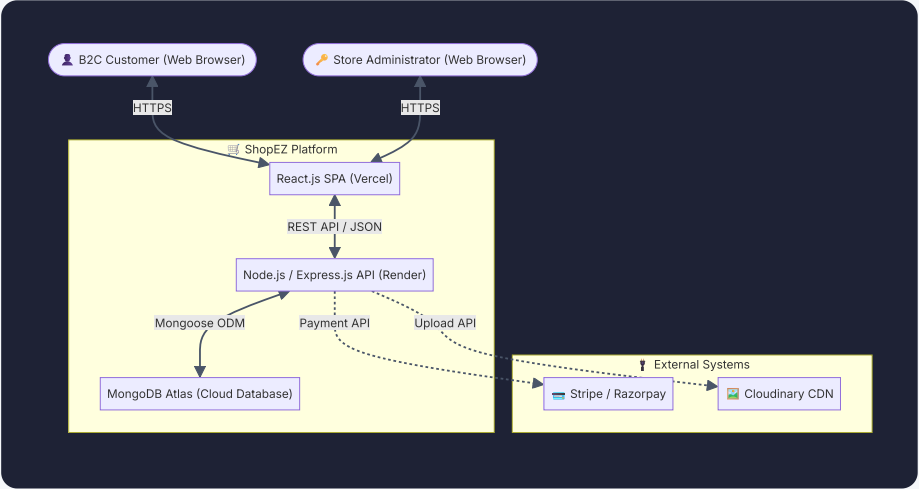


FIGURE 8.1 — C4 MODEL LEVEL 1: SYSTEM CONTEXT DIAGRAM

## 2. Solution Parameters

| S.No. | Parameter                                    | Detailed Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Problem Statement</b>                     | Customers require a secure, fast, and intuitive platform to discover and purchase items online. Administrators require a centralised, efficient tool to manage product inventory, monitor sales performance, and fulfil orders. The core gap is the absence of a unified, modern, customisable e-commerce solution accessible without prohibitive SaaS subscription costs.                                                               |
| 2     | <b>Solution Description</b>                  | A fully responsive Single Page Application (SPA) built on the MERN stack. The platform provides: (i) a customer-facing storefront with robust product discovery, persistent cart, and guided checkout; and (ii) an administrator panel with inventory management, order fulfilment, Cloudinary-powered image uploads, and sales analytics dashboards. Payment processing is delegated to PCI-DSS compliant gateways (Stripe / Razorpay). |
| 3     | <b>Novelty / Uniqueness</b>                  | Decoupled architecture with independently deployable frontend and backend; MongoDB-backed cross-device cart persistence; Cloudinary dynamic image transformations on CDN delivery; stateless JWT authentication enabling horizontal API scaling.                                                                                                                                                                                         |
| 4     | <b>Social Impact / Customer Satisfaction</b> | Delivers a reliable, transparent purchasing process building consumer trust through secure payment integration and clear order tracking. Empowers small business operators with accessible analytics, enabling data-                                                                                                                                                                                                                     |

| S.No. | Parameter                             | Detailed Description                                                                                                                                                                                                                       |
|-------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                       | driven inventory decisions and improved customer service.                                                                                                                                                                                  |
| 5     | <b>Business Model (Revenue Model)</b> | Primary: direct sales margin on products.<br>Secondary: premium product placement / featured listing slots; percentage-based commission for future multi-vendor configuration; shipping and handling fee collection on high-volume orders. |

### 3. High-Level Feature Summary

| Feature Category                     | Key Capabilities                                                            |
|--------------------------------------|-----------------------------------------------------------------------------|
| <b>Authentication &amp; Security</b> | JWT + Bcrypt; HTTP-only cookie sessions; RBAC (User / Admin)                |
| <b>Product Discovery</b>             | Full-text search; Category, Price, Rating filters; Sorting options          |
| <b>Shopping Experience</b>           | Persistent cross-device cart; Real-time total calculation; Stock validation |
| <b>Checkout &amp; Payment</b>        | 3-step wizard; Stripe / Razorpay integration; Invoice generation            |
| <b>Order Management</b>              | Order history (customer); Fulfilment status tracking (admin)                |
| <b>Product Reviews</b>               | Verified purchase reviews; Aggregate rating recalculation                   |
| <b>Admin Operations</b>              | Product CRUD; Category management; Cloudinary image upload                  |
| <b>Admin Analytics</b>               | Revenue charts; Order volume metrics; User growth indicators                |

